

SERVER SCRIPT

ALL IN ONE BOM SCRIPT

Script Type: DocType Event
Reference Document Type: Quotation
Event Frequency: All
DocType Event: Before Save

Script

```
# =====  
# FACTORY SMART ITEM & BOM MANAGER (SANDBOX FIX)  
# EVENT : Before Save (Quotation / Sales Order)  
# =====  
  
PROCESS_CODE = "100"  
PLACEHOLDER_ITEM = "CUSTOM-FABRIC"  
  
# ----- QUALITY CONFIG -----  
quality_config = {  
    "PREMIUM": "100", "PLATINUM": "101", "SUPER PLATINUM": "102",  
    "GOLD": "103", "SILVER": "104", "BRONZE": "105",  
    "CLASSIC": "106", "SUPER CLASSIC": "107", "LIFESTYLE": "108",  
    "ECO SPECIAL": "109", "ECO GREEN": "110", "SUPER ECO": "111",  
    "ULTRA": "112", "DELUXE": "113", "UV": "114"  
}  
quality_rev = {v:k for k,v in quality_config.items()}  
  
# ----- COLOR CONFIG -----  
color_config = {  
    "BRIGHT WHITE": "001",  
    "MILKY WHITE": "002",  
    "SUPER WHITE": "003",  
    "SUNSHINE WHITE": "004",  
    "BLEACH WHITE 1.0": "005",  
    "BLEACH WHITE 2.0": "006",  
    "BRIGHT IVORY": "101",  
    "IVORY 2.0": "102",  
    "CREAM 1.0": "121",  
    "LEMON YELLOW": "142",  
    "LEMON YELLOW 2.0": "143",  
    "LEMON YELLOW 4.0": "145",  
    "GOLDEN YELLOW": "161",  
    "GOLDEN YELLOW 5.0": "166",  
    "BRIGHT ORANGE": "181",  
    "DARK ORANGE": "182",  
    "ORANGE 1.0": "183",
```

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```
"BABY PINK": "201",
"DARK PINK": "204",
"PINK 4.0": "206",
"CRIMSON RED": "221",
"RED 1.0": "222",
"LIGHT MAROON": "241",
"RED 2.0": "242",
"DARK MAROON": "251",
"MEDICAL BLUE": "261",
"BLUE 3.0": "283",
"BLUE 5.0": "301",
"PEACOCK BLUE": "302",
"ROYAL BLUE": "322",
"NAVY BLUE": "347",
"VIOLET": "361",
"PARROT GREEN": "401",
"RELIANCE GREEN": "421",
"SEA GREEN": "446",
"LIGHT GREY": "464",
"DARK GREY": "465",
"CHOCOLATE BROWN": "487",
"LIGHT BEIGE": "501",
"DARK BEIGE": "521",
"BLACK": "542",
"WHITE MIX": "600",
"BLACK MIX": "601",
"COLOR MIX": "602",
"BEIGE MIX": "603"
}
color_rev = {v:k for k,v in color_config.items()}

# ----- LDR MAPPING -----
MB_LDR_MAP = {
    "MB - 1001001": 1.2, "MB - 1001002": 0.8, "MB - 1001003": 1.0, "MB - 1001004": 1.1,
    "MB - 1001005": 0.0, "MB - 1001006": 0.0, "MB - 1001101": 1.3, "MB - 1001121": 1.0,
    "MB - 1001122": 2.0, "MB - 1001123": 1.8, "MB - 1001124": 2.0, "MB - 1001125": 2.0,
    "MB - 1001126": 2.0, "MB - 1001141": 0.8, "MB - 1001142": 1.8, "MB - 1001143": 1.6,
    "MB - 1001144": 2.0, "MB - 1001161": 1.8, "MB - 1001162": 2.0, "MB - 1001163": 2.0,
    "MB - 1001164": 2.0, "MB - 1001165": 2.0, "MB - 1001166": 2.0, "MB - 1001181": 2.0,
    "MB - 1001182": 2.2, "MB - 1001183": 1.4, "MB - 1001184": 3.0, "MB - 1001185": 2.0,
    "MB - 1001187": 0.5, "MB - 1001201": 1.8, "MB - 1001202": 2.0, "MB - 1001203": 2.0,
    "MB - 1001204": 1.8, "MB - 1001205": 1.8, "MB - 1001206": 1.6, "MB - 1001207": 2.0,
    "MB - 1001208": 2.5, "MB - 1001209": 2.0, "MB - 1001221": 3.0, "MB - 1001222": 2.0,
    "MB - 1001241": 2.0, "MB - 1001251": 2.0, "MB - 1001252": 2.0, "MB - 1001253": 2.5,
    "MB - 1001261": 1.3, "MB - 1001262": 2.0, "MB - 1001263": 1.6, "MB - 1001281": 2.0,
    "MB - 1001282": 1.6, "MB - 1001283": 1.8, "MB - 1001284": 1.0, "MB - 1001285": 2.0,
    "MB - 1001286": 2.0, "MB - 1001287": 2.0, "MB - 1001201": 1.6, "MB - 1001202": 2.0
```

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```
MB - 1001200": 3.0, "MB - 1001207": 2.0, "MB - 1001301": 1.0, "MB - 1001302": 2.0,
"MB - 1001303": 1.0, "MB - 1001321": 2.0, "MB - 1001322": 2.0, "MB - 1001323": 2.25,
"MB - 1001344": 2.0, "MB - 1001345": 2.0, "MB - 1001346": 2.0, "MB - 1001347": 2.0,
"MB - 1001361": 1.8, "MB - 1001362": 3.5, "MB - 1001363": 2.0, "MB - 1001364": 2.0,
"MB - 1001365": 2.0, "MB - 1001381": 1.0, "MB - 1001401": 2.0, "MB - 1001402": 2.0,
"MB - 1001421": 1.8, "MB - 1001422": 1.6, "MB - 1001441": 2.0, "MB - 1001442": 2.0,
"MB - 1001443": 2.0, "MB - 1001444": 1.5, "MB - 1001445": 2.95, "MB - 1001446": 2.1,
"MB - 1001447": 2.0, "MB - 1001448": 2.0, "MB - 1001449": 3.0, "MB - 1001451": 2.0,
"MB - 1001452": 2.0, "MB - 1001453": 2.5, "MB - 1001461": 0.4, "MB - 1001462": 2.0,
"MB - 1001463": 2.0, "MB - 1001464": 2.0, "MB - 1001465": 2.0, "MB - 1001466": 2.0,
"MB - 1001481": 2.0, "MB - 1001482": 2.0, "MB - 1001483": 2.0, "MB - 1001484": 2.0,
"MB - 1001485": 3.0, "MB - 1001486": 1.4, "MB - 1001487": 2.0, "MB - 1001488": 2.0,
"MB - 1001501": 1.8, "MB - 1001502": 2.0, "MB - 1001503": 2.0, "MB - 1001504": 2.0,
"MB - 1001505": 1.6, "MB - 1001521": 2.0, "MB - 1001522": 1.2, "MB - 1001523": 2.0,
"MB - 1001541": 2.0, "MB - 1001542": 3.0
}

# ----- MIXING RATIOS -----
mix_map = {
    "DELUXE":      [35, 100, 1.0, 0.5],
    "PREMIUM":     [100, 0, 0.5, 0.2],
    "GOLD":        [100, 25, 0.5, 0.2],
    "BRONZE":      [100, 35, 0.2, 0.2],
    "CLASSIC":     [80, 40, 0.5, 0.2],
    "SUPER ECO":   [100, 30, 0.5, 0.2],
    "PLATINUM":    [100, 10, 0.5, 0.2],
    "ECO SPECIAL": [55, 60, 0.5, 0.2],
    "ECO GREEN":   [60, 75, 1.0, 0.5],
    "LIFESTYLE":   [65, 50, 0.5, 0.2],
    "SILVER":      [100, 30, 0.5, 0.2],
    "SUPER PLATINUM": [100, 15, 0.5, 0.2],
    "SUPER CLASSIC": [75, 50, 0.5, 0.2],
    "ULTRA":       [40, 75, 1.0, 0.5]
}

# =====
# MAIN LOOP
# =====
if doc.get("items"):
    for row in doc.items:

        # =====
        #  EXISTING ITEM → AUTO FILL AFTER SAVE
        # =====
        if (row.item_code and row.item_code != PLACEHOLDER_ITEM
            and row.item_code.startswith(PROCESS_CODE) and len(row.item_code) == 16):
            try:
```

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```
code = row.item_code
row.custom_quality = quality_rev.get(code[3:6])
row.custom_color = color_rev.get(code[6:9])
row.custom_gsm = int(code[9:12])
row.custom_width_inch = round(int(code[12:16]) / 25.4, 1)
except Exception:
    pass

# =====
# CUSTOM FABRIC → CREATE ITEM & BOM
# =====
if row.item_code != PLACEHOLDER_ITEM:
    continue

q = str(row.custom_quality or "").strip().upper()
c = str(row.custom_color or "").strip().upper()

if c == "WHITE": c = "BRIGHT WHITE"
row.custom_color = c

if not (q and c and row.custom_gsm and row.custom_width_inch):
    continue

q_code = quality_config.get(q, "105")
c_code = color_config.get(c, "000")

width = float(row.custom_width_inch)
mm = int(5 * round((width * 25.4)/5))

gsm_code = str(int(row.custom_gsm)).zfill(3)
mm_code = str(mm).zfill(4)

final_item_code = f"{PROCESS_CODE}{q_code}{c_code}{gsm_code}{mm_code}"
final_item_name = f"NON WOVEN FABRIC {q} {c} {row.custom_gsm} GSM W - {width}" ( {mm} MM )"

# ----- HSN -----
gsm = int(row.custom_gsm)
hsn = "56031100"
if 25 <= gsm <= 70: hsn = "56031200"
elif 71 <= gsm <= 150: hsn = "56031300"
elif gsm > 150: hsn = "56031400"

# ----- CREATE ITEM -----
if not frappe.db.exists("Item", final_item_code):

    item = frappe.new_doc("Item")
```

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```
item.item_code = final_item_code
item.item_name = final_item_name
item.item_group = "Products"
item.stock_uom = "Kg"
item.gst_hsn_code = hsn

item.append("item_defaults",{
    "company": doc.company,
    "default_warehouse": "Finished Goods - JSB-1ZT"
})
item.insert(ignore_permissions=True)

# =====
#  AUTO-CREATE & SUBMIT BOM INSTANTLY
# =====
try:
    # Predict Masterbatch Item Code based on color config mapping
    mb_item_code = f"MB - 1001{c_code}"

    # Fetch LDR from your dictionary (default to 1.1 if not found)
    ldr_percent = float(MB_LDR_MAP.get(mb_item_code, 1.1))

    # Check if Masterbatch exists. If it doesn't, WARN THE USER and skip BOM!
    if not frappe.db.exists("Item", mb_item_code):
        frappe.msgprint(f"⚠ <b>BOM Skipped:</b> The Masterbatch item <b>{mb_item_code}</b> does not exist in your Item List. Create it first, then create the BOM manually.")
    else:
        ITEM_PP = "PP - 1002001"
        ITEM_FL = "FL - 1003011"
        ITEM_PPA = "SA - 1004001"
        ITEM_ANTI = "SA - 1004002"

        # Make sure raw materials exist before building BOM
        missing_rms = []
        for rm in [ITEM_PP, ITEM_PPA, ITEM_ANTI]:
            if not frappe.db.exists("Item", rm):
                missing_rms.append(rm)

        if len(missing_rms) > 0:
            frappe.msgprint(f"⚠ <b>BOM Skipped:</b> Missing standard Raw Materials in Item List: {' '.join(missing_rms)}")
        else:
            # Get mixing ratios
            ratios = mix_map.get(q, [100, 35, 0.2, 0.2])

            # THE FIX: Assign line by line instead of sequence unpacking
            parts_on = ratios[0]
```

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```
parts_fl = ratios[1]
parts_ppa = ratios[2]
parts_anti = ratios[3]

# Mathematical calculation exactly matching your API
base_tot = parts_pp + parts_fl
mb_weight = (base_tot * ldr_percent) / 100.0
total_batch_weight = base_tot + parts_ppa + parts_anti + mb_weight
one_mixing_factor = 1.0 / total_batch_weight

t_pp = round(parts_pp * one_mixing_factor, 3)
t_fl = round(parts_fl * one_mixing_factor, 3)
t_ppa = round(parts_ppa * one_mixing_factor, 3)
t_anti = round(parts_anti * one_mixing_factor, 3)
t_mb = round(mb_weight * one_mixing_factor, 3)

# Create and submit the BOM Document
bom = frappe.new_doc("BOM")
bom.item = final_item_code
bom.quantity = 1.0
bom.is_default = 1
bom.is_active = 1
bom.currency = "INR"
bom.rm_cost_as_per = "Valuation Rate"

# Assign custom LDR field
bom.custom_ldr = ldr_percent

bom.append("items", {"item_code": ITEM_PP, "qty": t_pp, "uom": "Kg"})
if t_fl > 0:
    # Only add filler if it exists in system
    if frappe.db.exists("Item", ITEM_FL):
        bom.append("items", {"item_code": ITEM_FL, "qty": t_fl, "uom": "Kg"})
    else:
        frappe.msgprint(f"⚠ <b>Warning:</b> Missing filler item {ITEM_FL}. BOM created without filler")

bom.append("items", {"item_code": ITEM_PPA, "qty": t_ppa, "uom": "Kg"})
bom.append("items", {"item_code": ITEM_ANTI, "qty": t_anti, "uom": "Kg"})
bom.append("items", {"item_code": mb_item_code, "qty": t_mb, "uom": "Kg"})

bom.insert(ignore_permissions=True)
bom.submit()
frappe.msgprint(f"✅ <b>Success:</b> Item & BOM created for {final_item_code}")
```

except Exception as e:

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```
frappe.msgprint(f"⚠ <b>BOM Creation Failed for {final_item_code}</b> {str(e)}")
frappe.log_error(f"Auto BOM Creation Failed for {final_item_code}", str(e))

# ----- SWAP ITEM IN CURRENT DOC -----
row.item_code = final_item_code
row.item_name = final_item_name
row.description = final_item_name

item_doc = frappe.get_doc("Item", final_item_code)

row.gst_hsn_code = item_doc.gst_hsn_code
row.uom = item_doc.stock_uom
row.stock_uom = item_doc.stock_uom
row.conversion_factor = 1#=====
# FACTORY SMART ITEM & BOM MANAGER (SANDBOX FIX)
# EVENT : Before Save (Quotation / Sales Order)
# =====

PROCESS_CODE = "100"
PLACEHOLDER_ITEM = "CUSTOM-FABRIC"

# ----- QUALITY CONFIG -----
quality_config = {
    "PREMIUM": "100", "PLATINUM": "101", "SUPER PLATINUM": "102",
    "GOLD": "103", "SILVER": "104", "BRONZE": "105",
    "CLASSIC": "106", "SUPER CLASSIC": "107", "LIFESTYLE": "108",
    "ECO SPECIAL": "109", "ECO GREEN": "110", "SUPER ECO": "111",
    "ULTRA": "112", "DELUXE": "113", "UV": "114"
}
quality_rev = {v:k for k,v in quality_config.items()}

# ----- COLOR CONFIG -----
color_config = {
    "BRIGHT WHITE": "001",
    "MILKY WHITE": "002",
    "SUPER WHITE": "003",
    "SUNSHINE WHITE": "004",
    "BLEACH WHITE 1.0": "005",
    "BLEACH WHITE 2.0": "006",
    "BRIGHT IVORY": "101",
    "IVORY 2.0": "102",
    "CREAM 1.0": "121",
    "LEMON YELLOW": "142",
    "LEMON YELLOW 2.0": "143",
    "LEMON YELLOW 4.0": "145",
    "GOLDEN YELLOW": "161",
    "GOLDEN YELLOW 5.0": "166"
```

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```
GOLDEN YELLOW 3.0": "180",
"BRIGHT ORANGE": "181",
"DARK ORANGE": "182",
"ORANGE 1.0": "183",
"BABY PINK": "201",
"DARK PINK": "204",
"PINK 4.0": "206",
"CRIMSON RED": "221",
"RED 1.0": "222",
"LIGHT MAROON": "241",
"RED 2.0": "242",
"DARK MAROON": "251",
"MEDICAL BLUE": "261",
"BLUE 3.0": "283",
"BLUE 5.0": "301",
"PEACOCK BLUE": "302",
"ROYAL BLUE": "322",
"NAVY BLUE": "347",
"VIOLET": "361",
"PARROT GREEN": "401",
"RELIANCE GREEN": "421",
"SEA GREEN": "446",
"LIGHT GREY": "464",
"DARK GREY": "465",
"CHOCOLATE BROWN": "487",
"LIGHT BEIGE": "501",
"DARK BEIGE": "521",
"BLACK": "542",
"WHITE MIX": "600",
"BLACK MIX": "601",
"COLOR MIX": "602",
"BEIGE MIX": "603"
}
color_rev = {v:k for k,v in color_config.items()}

# ----- LDR MAPPING -----
MB_LDR_MAP = {
    "MB - 1001001": 1.2, "MB - 1001002": 0.8, "MB - 1001003": 1.0, "MB - 1001004": 1.1,
    "MB - 1001005": 0.0, "MB - 1001006": 0.0, "MB - 1001101": 1.3, "MB - 1001121": 1.0,
    "MB - 1001122": 2.0, "MB - 1001123": 1.8, "MB - 1001124": 2.0, "MB - 1001125": 2.0,
    "MB - 1001126": 2.0, "MB - 1001141": 0.8, "MB - 1001142": 1.8, "MB - 1001143": 1.6,
    "MB - 1001144": 2.0, "MB - 1001161": 1.8, "MB - 1001162": 2.0, "MB - 1001163": 2.0,
    "MB - 1001164": 2.0, "MB - 1001165": 2.0, "MB - 1001166": 2.0, "MB - 1001181": 2.0,
    "MB - 1001182": 2.2, "MB - 1001183": 1.4, "MB - 1001184": 3.0, "MB - 1001185": 2.0,
    "MB - 1001187": 0.5, "MB - 1001201": 1.8, "MB - 1001202": 2.0, "MB - 1001203": 2.0,
    "MB - 1001204": 1.8, "MB - 1001205": 1.8, "MB - 1001206": 1.6, "MB - 1001207": 2.0,
    "MB - 1001208": 2.5, "MB - 1001209": 2.0, "MB - 1001221": 3.0, "MB - 1001222": 2.0,
```


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```
"MB - 1001241": 2.0, "MB - 1001251": 2.0, "MB - 1001252": 2.0, "MB - 1001253": 2.5,
"MB - 1001261": 1.3, "MB - 1001262": 2.0, "MB - 1001263": 1.6, "MB - 1001281": 2.0,
"MB - 1001282": 1.6, "MB - 1001283": 1.8, "MB - 1001284": 1.0, "MB - 1001285": 2.0,
"MB - 1001286": 3.0, "MB - 1001287": 2.3, "MB - 1001301": 1.6, "MB - 1001302": 2.0,
"MB - 1001303": 1.0, "MB - 1001321": 2.0, "MB - 1001322": 2.0, "MB - 1001323": 2.25,
"MB - 1001344": 2.0, "MB - 1001345": 2.0, "MB - 1001346": 2.0, "MB - 1001347": 2.0,
"MB - 1001361": 1.8, "MB - 1001362": 3.5, "MB - 1001363": 2.0, "MB - 1001364": 2.0,
"MB - 1001365": 2.0, "MB - 1001381": 1.0, "MB - 1001401": 2.0, "MB - 1001402": 2.0,
"MB - 1001421": 1.8, "MB - 1001422": 1.6, "MB - 1001441": 2.0, "MB - 1001442": 2.0,
"MB - 1001443": 2.0, "MB - 1001444": 1.5, "MB - 1001445": 2.95, "MB - 1001446": 2.1,
"MB - 1001447": 2.0, "MB - 1001448": 2.0, "MB - 1001449": 3.0, "MB - 1001451": 2.0,
"MB - 1001452": 2.0, "MB - 1001453": 2.5, "MB - 1001461": 0.4, "MB - 1001462": 2.0,
"MB - 1001463": 2.0, "MB - 1001464": 2.0, "MB - 1001465": 2.0, "MB - 1001466": 2.0,
"MB - 1001481": 2.0, "MB - 1001482": 2.0, "MB - 1001483": 2.0, "MB - 1001484": 2.0,
"MB - 1001485": 3.0, "MB - 1001486": 1.4, "MB - 1001487": 2.0, "MB - 1001488": 2.0,
"MB - 1001501": 1.8, "MB - 1001502": 2.0, "MB - 1001503": 2.0, "MB - 1001504": 2.0,
"MB - 1001505": 1.6, "MB - 1001521": 2.0, "MB - 1001522": 1.2, "MB - 1001523": 2.0,
"MB - 1001541": 2.0, "MB - 1001542": 3.0
}
```

```
# ----- MIXING RATIOS -----
```

```
mix_map = {
  "DELUXE":    [35, 100, 1.0, 0.5],
  "PREMIUM":   [100, 0, 0.5, 0.2],
  "GOLD":      [100, 25, 0.5, 0.2],
  "BRONZE":    [100, 35, 0.2, 0.2],
  "CLASSIC":   [80, 40, 0.5, 0.2],
  "SUPER ECO": [100, 30, 0.5, 0.2],
  "PLATINUM":  [100, 10, 0.5, 0.2],
  "ECO SPECIAL": [55, 60, 0.5, 0.2],
  "ECO GREEN": [60, 75, 1.0, 0.5],
  "LIFESTYLE": [65, 50, 0.5, 0.2],
  "SILVER":    [100, 30, 0.5, 0.2],
  "SUPER PLATINUM": [100, 15, 0.5, 0.2],
  "SUPER CLASSIC": [75, 50, 0.5, 0.2],
  "ULTRA":     [40, 75, 1.0, 0.5]
}
```

```
# =====
```

```
# MAIN LOOP
```

```
# =====
```

```
if doc.get("items"):
    for row in doc.items:
```

```
    # =====
```

```
    #  EXISTING ITEM → AUTO FILL AFTER SAVE
```

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```
# =====
if (row.item_code and row.item_code != PLACEHOLDER_ITEM
    and row.item_code.startswith(PROCESS_CODE) and len(row.item_code) == 16):
    try:
        code = row.item_code
        row.custom_quality = quality_rev.get(code[3:6])
        row.custom_color = color_rev.get(code[6:9])
        row.custom_gsm = int(code[9:12])
        row.custom_width_inch = round(int(code[12:16]) / 25.4, 1)
    except Exception:
        pass

# =====
# CUSTOM FABRIC → CREATE ITEM & BOM
# =====
if row.item_code != PLACEHOLDER_ITEM:
    continue

q = str(row.custom_quality or "").strip().upper()
c = str(row.custom_color or "").strip().upper()

if c == "WHITE": c = "BRIGHT WHITE"
row.custom_color = c

if not (q and c and row.custom_gsm and row.custom_width_inch):
    continue

q_code = quality_config.get(q,"105")
c_code = color_config.get(c,"000")

width = float(row.custom_width_inch)
mm = int(5 * round((width * 25.4)/5))

gsm_code = str(int(row.custom_gsm)).zfill(3)
mm_code = str(mm).zfill(4)

final_item_code = f"{PROCESS_CODE}{q_code}{c_code}{gsm_code}{mm_code}"
final_item_name = f"NON WOVEN FABRIC {q} {c} {row.custom_gsm} GSM W - {width}" ( {mm} MM )"

# ----- HSN -----
gsm = int(row.custom_gsm)
hsn = "56031100"
if 25 <= gsm <= 70: hsn = "56031200"
elif 71 <= gsm <= 150: hsn = "56031300"
elif gsm > 150: hsn = "56031400"

# ----- CREATE ITEM -----
```

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```
if not frappe.db.exists("Item", final_item_code):

    item = frappe.new_doc("Item")
    item.item_code = final_item_code
    item.item_name = final_item_name
    item.item_group = "Products"
    item.stock_uom = "Kg"
    item.gst_hsn_code = hsn

    item.append("item_defaults",{
        "company": doc.company,
        "default_warehouse": "Finished Goods - JSB-1ZT"
    })
    item.insert(ignore_permissions=True)

# =====
#  AUTO-CREATE & SUBMIT BOM INSTANTLY
# =====
try:
    # Predict Masterbatch Item Code based on color config mapping
    mb_item_code = f"MB - 1001{c_code}"

    # Fetch LDR from your dictionary (default to 1.1 if not found)
    ldr_percent = float(MB_LDR_MAP.get(mb_item_code, 1.1))

    # Check if Masterbatch exists. If it doesn't, WARN THE USER and skip BOM!
    if not frappe.db.exists("Item", mb_item_code):
        frappe.msgprint(f"⚠ <b>BOM Skipped:</b> The Masterbatch item <b>{mb_item_code}</b> does not exist in your Item List. Create it first, then create the BOM manually.")
    else:
        ITEM_PP = "PP - 1002001"
        ITEM_FL = "FL - 1003011"
        ITEM_PPA = "SA - 1004001"
        ITEM_ANTI = "SA - 1004002"

        # Make sure raw materials exist before building BOM
        missing_rms = []
        for rm in [ITEM_PP, ITEM_PPA, ITEM_ANTI]:
            if not frappe.db.exists("Item", rm):
                missing_rms.append(rm)

        if len(missing_rms) > 0:
            frappe.msgprint(f"⚠ <b>BOM Skipped:</b> Missing standard Raw Materials in Item List: {' '.join(missing_rms)}")
        else:
            # Get mixing ratios
```

SERVER SCRIPT

ALL IN ONE BOM SCRIPT

```
ratios = mix_map.get(q, [100, 35, 0.2, 0.2])

# THE FIX: Assign line by line instead of sequence unpacking
parts_pp = ratios[0]
parts_fl = ratios[1]
parts_ppa = ratios[2]
parts_anti = ratios[3]

# Mathematical calculation exactly matching your API
base_tot = parts_pp + parts_fl
mb_weight = (base_tot * ldr_percent) / 100.0
total_batch_weight = base_tot + parts_ppa + parts_anti + mb_weight
one_mixing_factor = 1.0 / total_batch_weight

t_pp = round(parts_pp * one_mixing_factor, 3)
t_fl = round(parts_fl * one_mixing_factor, 3)
t_ppa = round(parts_ppa * one_mixing_factor, 3)
t_anti = round(parts_anti * one_mixing_factor, 3)
t_mb = round(mb_weight * one_mixing_factor, 3)

# Create and submit the BOM Document
bom = frappe.new_doc("BOM")
bom.item = final_item_code
bom.quantity = 1.0
bom.is_default = 1
bom.is_active = 1
bom.currency = "INR"
bom.rm_cost_as_per = "Valuation Rate"

# Assign custom LDR field
bom.custom_ldr_ = ldr_percent

bom.append("items", {"item_code": ITEM_PP, "qty": t_pp, "uom": "Kg"})
if t_fl > 0:
    # Only add filler if it exists in system
    if frappe.db.exists("Item", ITEM_FL):
        bom.append("items", {"item_code": ITEM_FL, "qty": t_fl, "uom": "Kg"})
    else:
        frappe.msgprint(f"⚠ <b>Warning:</b> Missing filler item {ITEM_FL}. BOM created without filler")

bom.append("items", {"item_code": ITEM_PPA, "qty": t_ppa, "uom": "Kg"})
bom.append("items", {"item_code": ITEM_ANTI, "qty": t_anti, "uom": "Kg"})
bom.append("items", {"item_code": mb_item_code, "qty": t_mb, "uom": "Kg"})

bom.insert(ignore_permissions=True)
```

SERVER SCRIPT

ALL IN ONE BOM SCRIPT

```
    bom.submit()
    frappe.msgprint(f"✅ <b>Success:</b> Item & BOM created for {final_item_code}")

except Exception as e:
    frappe.msgprint(f"⚠️ <b>BOM Creation Failed for {final_item_code}:</b> {str(e)}")
    frappe.log_error(f"Auto BOM Creation Failed for {final_item_code}", str(e))

# ----- SWAP ITEM IN CURRENT DOC -----
row.item_code = final_item_code
row.item_name = final_item_name
row.description = final_item_name

item_doc = frappe.get_doc("Item", final_item_code)

row.gst_hsn_code = item_doc.gst_hsn_code
row.uom = item_doc.stock_uom
row.stock_uom = item_doc.stock_uom
row.conversion_factor = 1
```

Request Limit:

5

Time Window (Seconds):

86400